Enzyme reactions III

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Biomatematica 2

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Competitive inhibition

Competitive enzyme inhibition is a type of reaction in which a substance, called inhibitor (I), competes with the substrate to bind to the enzyme.

The diagram of reaction is the following:

$$\mathrm{S} + \mathrm{E} \underset{k_{-1}}{\overset{k_1}{\rightleftarrows}} \mathrm{C}_1 \overset{k_2}{\rightarrow} \mathrm{P} + \mathrm{E}$$

$$\mathrm{I} + \mathrm{E} \underset{k_{-3}}{\overset{k_3}{\rightleftarrows}} \mathrm{C}_2$$

Competitive inhibition: ODE system

Denoting by i the concentration of the inhibitor and applying the law of mass action, we obtain the ODE system:

$$\begin{cases} \dfrac{ds}{dt} = k_{-1}c_1 - k_1 es & s(0) = s_0 \\ \dfrac{de}{dt} = (k_{-1} + k_2)c_1 - k_1 es + k_{-3}c_2 - k_3 ei & e(0) = e_0 \\ \dfrac{di}{dt} = -k_3 ei + k_{-3}c_2 & i(0) = i_0 \\ \dfrac{dc_1}{dt} = k_1 es - (k_{-1} + k_2)c_1 & c_1(0) = 0 \\ \dfrac{dc_2}{dt} = k_3 ei - k_{-3}c_2 & c_2(0) = 0 \\ \dfrac{dp}{dt} = k_2 c & p(0) = 0 \end{cases} \tag{1}$$

Competitive inhibition: conservation properties

- Inhibitor conservation:

$$\frac{di}{dt} + \frac{dc_2}{dt} = 0 \Rightarrow i + c_2 = i_0;$$

- Enzyme conservation:

$$\frac{de}{dt} + \frac{dc_1}{dt} + \frac{dc_2}{dt} = 0 \Rightarrow e + c_1 + c_2 = e_0;$$

- Substrate conservation:

$$\frac{ds}{dt} + \frac{dc_1}{dt} + \frac{dp}{dt} = 0 \Rightarrow s + c_1 + p = s_0.$$

Competitive inhibition: the reduced system

By setting $e = e_0 - c_1 - c_2$ and eliminating the equation for the product p, the reduced system reads:

$$\begin{cases} \dfrac{ds}{dt} = -k_1 e_0 s + k_1 s c_2 + (k_1 s + k_{-1}) c_1 & s(0) = s_0 \\ \dfrac{di}{dt} = -k_3 e_0 i + k_3 i c_1 + (k_3 i + k_{-3}) c_2 & i(0) = i_0 \\ \dfrac{dc_1}{dt} = k_1 e_0 s - k_1 s c_2 - (k_1 s + k_{-1} + k_2) c_1 & c_1(0) = 0 \\ \dfrac{dc_2}{dt} = k_3 e_0 i - k_3 i c_1 - (k_3 i + k_{-3}) c_2 & c_2(0) = 0 \end{cases} \tag{2}$$

Competitive inhibition: quasi-steady state approximation

Assume that

$$\frac{dc_1}{dt} = \frac{dc_2}{dt} = 0.$$

Then, after setting $K_i = \frac{k_{-3}}{k_3}$ and $K_m = \frac{k_{-1}+k_2}{k_1}$, we obtain:

$$c_1 = \frac{e_0 s}{s + K_m\left(1 + \frac{i}{K_i}\right)} \qquad \text{and} \qquad c_2 = \frac{K_m e_0 i}{K_i\left(s + K_m\left(1 + \frac{i}{K_i}\right)\right)}$$

Thus, the reaction velocity in the quasi-steady state is given by:

$$v = \frac{dp}{dt} = k_2 c_1 = \frac{k_2 e_0 s}{s + K_m\left(1 + \frac{i}{K_i}\right)}.$$

Cooperativity

Suppose that the enzyme can bind two substrate molecules, so it can exist in one of the following three states:

- a free molecule E;
- a complex with one occupied binding site C_1;
- a complex with two occupied binding sites C_2.

The reaction diagram is:

$$S + E \underset{k_{-1}}{\overset{k_1}{\rightleftarrows}} C_1 \overset{k_2}{\rightarrow} P + E$$

$$S + C_1 \underset{k_{-3}}{\overset{k_3}{\rightleftarrows}} C_2 \overset{k_4}{\rightarrow} P + C_1$$

Cooperativity: ODE system

By applying the law of mass action, we obtain the ODE system:

$$\begin{cases} \dfrac{ds}{dt} = -k_1 es + k_{-1} c_1 - k_3 s c_1 + k_{-3} c_2 & s(0) = s_0 \\ \dfrac{de}{dt} = -k_1 es + (k_{-1} + k_2) c_1 & e(0) = e_0 \\ \dfrac{dc_1}{dt} = k_1 es - (k_{-1} + k_2) c_1 - k_3 s c_1 + (k_{-3} + k_4) c_2 & c_1(0) = 0 \\ \dfrac{dc_2}{dt} = k_3 s c_1 - (k_{-3} + k_4) c_2 & c_2(0) = 0 \\ \dfrac{dp}{dt} = k_2 c_1 + k_4 c_2 & p(0) = 0 \end{cases} \tag{3}$$

Cooperativity: conservation properties

- Enzyme conservation:

$$\frac{de}{dt} + \frac{dc_1}{dt} + \frac{dc_2}{dt} = 0 \Rightarrow e + c_1 + c_2 = e_0;$$

- Substrate conservation:

$$\frac{ds}{dt} + \frac{dc_1}{dt} + 2\frac{dc_2}{dt} + \frac{dp}{dt} = 0 \Rightarrow s + c_1 + 2c_2 + p = s_0.$$

Cooperativity: reduced system

By setting $e = e_0 - c_1 - c_2$ and eliminating the equation for the product p, the reduced system reads:

$$\begin{cases} \dfrac{ds}{dt} = -k_1 e_0 s + (k_1 - k_3)c_1 s + k_1 c_2 s + k_{-1} c_1 + k_{-3} c_2 \\ \dfrac{dc_1}{dt} = k_1 e_0 s - (k_1 + k_3)c_1 s - k_1 c_2 s - (k_{-1} + k_2)c_1 + (k_{-3} + k_4)c_2 \\ \dfrac{dc_2}{dt} = k_3 c_1 s - (k_{-3} + k_4)c_2 \end{cases} \tag{4}$$

with initial conditions $s(0) = s_0$, $c_1(0) = 0$, $c_2(0) = 0$.

Cooperativity: quasi-steady state approximation

Assume that

$$\frac{dc_1}{dt} = \frac{dc_2}{dt} = 0.$$

Then, after setting $K_c = \frac{k_{-3}+k_4}{k_3}$ and $K_m = \frac{k_{-1}+k_2}{k_1}$, we obtain:

$$c_1 = \frac{K_c e_0 s}{K_c s + s^2 + K_m K_c} \qquad \text{and} \qquad c_2 = \frac{e_0 s^2}{K_c s + s^2 + K_m K_c}$$

Thus, the reaction velocity in the quasi-steady state is given by:

$$v = \frac{dp}{dt} = k_2 c_1 + k_4 c_2 = \frac{(k_2 K_c + k_4 s) e_0 s}{K_c s + s^2 + K_m K_c}.$$

No cooperativity

The enzyme binding sites act independently and identically, then

$$k_1 = 2k_3 = 2k_+, \qquad 2k_{-1} = k_{-3} = 2k_-, \qquad 2k_2 = k_4,$$

where k_+ and k_- are the forward and backward reaction rates for the individual binding sites.

The factors 2 occur because two identical binding sites are involved in the reaction, doubling the amount of the reactant.

Thus the reaction velocity is given by the Michaelis-Menten formula

$$v = \frac{2k_2 e_0 (K+s)s}{K^2 + 2Ks + s^2} = \frac{2k_2 e_0 s}{K+s},$$

where $K = \frac{k_- + k_2}{k_+}$.

Positive cooperativity

Binding of the first substrate molecule is slow, but binding of the second is fast.

This can be modeled by letting

$$k_3 \to +\infty \qquad \text{and} \qquad k_1 \to 0,$$

while keeping $k_1 k_3$ constant.

As a consequence

$$K_c \to 0 \qquad \text{and} \qquad K_m \to +\infty,$$

whith $K_m K_c$ constant.

In this case, the reaction velocity is given by

$$v = \frac{k_4 e_0 s^2}{s^2 + K_m K_c}. \tag{5}$$

(5) is known as Hill equation.

Negative cooperativity

Binding of the first substrate molecule decreases the rate of subsequent binding.

This can be modeled by letting

$$k_3 \to 0 \qquad \text{and} \qquad k_1 \to +\infty,$$

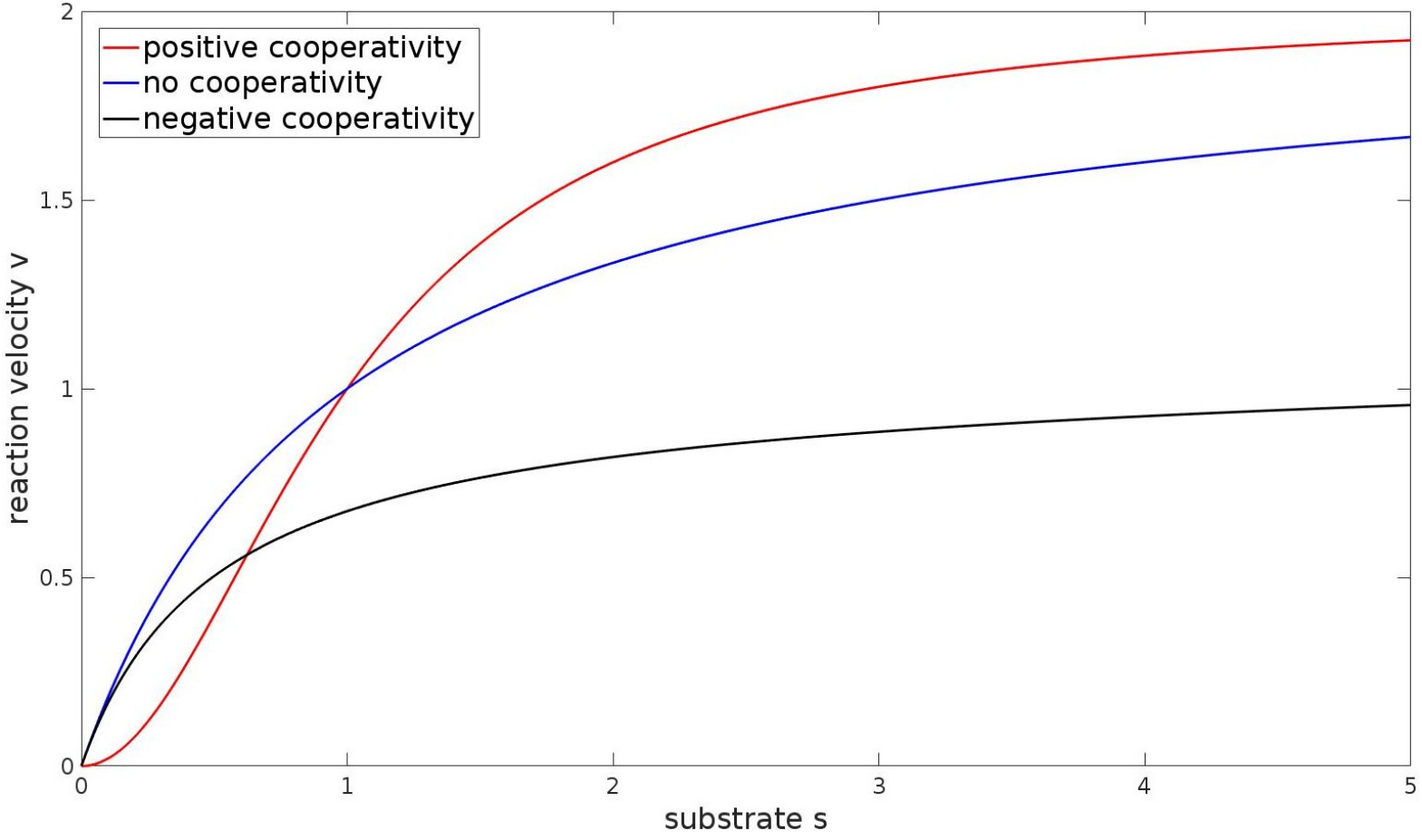
while keeping $k_1 k_3$ constant.

As a consequence

$$K_c \to +\infty \qquad \text{and} \qquad K_m \to 0,$$

whith $K_m K_c$ constant.

In this case, the reaction velocity is given by the Michaelis-Menten formula

$$v = \frac{k_2 e_0 s}{s + K_m}. \tag{6}$$

Plot of reaction velocity in the three cases